Applying Machine Learning to the Fuel Theft Problem on Pipelines

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Abstract: Fuel theft of oil pipelines is a concern faced by various countries and oil and gas companies due to its impact on the environment and the safety of neighboring communities. Oil pipeline monitoring and inspection with the support of alerts is essential to preventing suspected fuel-theft events and mitigating risks. Monitoring systems trigger alerts, and patrols are sent to confirm the occurrence of illegal tapping. However, various signals can be activated briefly, and the correct prioritization is essential to identifying illegal tapping as quickly as possible. This work aims to use machine-learning techniques to develop a predictive model capable of forecasting the probability of an event resulting in illegal tapping and understanding the factors that influence the occurrence. A Brazilian oil and gas transportation company provided data from a monitoring system supervised from January 2019 to August 2021. Five machine-learning algorithms were used in this study: Logistic Regression, Random Forest, XGBoost, Catboost, and Multilayer Perceptron. The Random Forest obtained the best results in classifying alerts associated (or not) with an illegal tapping, showing accuracy, specificity, and sensitivities of 76.8%, 68.3%, and 100%, respectively. In this problem, specificity implies reducing the sending of patrols to the field in 68.3% of circumstances, and sensitivity means that the model is a good predictor for positive cases. As for the external validation, the model also performed well, with an accuracy and specificity of 61%. The factors that most highly influenced illegal tapping occurrences were the alert duration, previous events in the same area, and events during the night. **DOI: 10.1061/JPSEA2.PSENG-1374.** *© 2023 American Society of Civil Engineers.*

Practical Applications: The problem of fuel theft of oil pipelines affects several countries worldwide and is a major concern due to its impact on the environment and the safety of neighboring communities. Oil pipeline surveillance through monitoring systems is important to prevent suspected fuel-theft events and mitigate risks. However, many alerts can be triggered in a brief period, and their correct prioritization is essential to identifying illegal tapping as quickly as possible. This work used machine-learning techniques to develop a predictive model capable of forecasting the probability of an event resulting in illegal tapping. Moreover, it has identified the factors that most influence a fuel-theft occurrence: alert duration, previous events in the same area, and the night period. The developed model can be used to assist companies in prioritizing alerts and improving the efficiency of patrol responses, contributing to combat, and reducing fuel theft.

Author keywords: Machine learning; Fuel theft; Illegal tapping; Oil pipelines; Data science.

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Introduction

Brazil is one of the largest oil producers globally, and the transport of this fossil fuel and its derivatives is partially carried out through oil and gas pipelines. According to ANP (2020), in 2019, the country had 530 pipelines used for the transportation of oil, derivatives, natural gas, and ethanol, totaling 21,600 km of pipeline. Pipelines represent the most efficient mode of transport and movement of oil and fuel and are also safe if there is no third-party intervention. However, one of the biggest problems carriers face is fuel theft from pipelines. According to Petrobras (2020), 201 theft (or attempted) cases were reported in 2020, preceded by 203 cases in 2019 and 261 in 2018. This unlawful practice, also known as illegal tapping, presents a risk to the environment due to the possibility of leaks, contamination of soil and rivers, fires, or explosions, and may impact communities near the pipelines. In addition, the impact of such crimes can reach more than US$40 million for Petrobras annually (Veja 2019).

The fuel theft problem impacts Brazil and several countries worldwide, including Mexico and Nigeria. According to GFI (2017), a conservative estimate of crude oil theft's worth is at least US $5.2 billion to US$11.9 billion annually as of 2015. This estimate

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is based only on six countries: Colombia, Indonesia, Mexico, Syria, Russia, and Nigeria, which all present high levels of crude oil theft. According to Yale University (2020), this practice finances several criminal organizations and is thus a primary global security concern.

Monitoring and preventing the suspected fuel theft events in pipelines is essential to mitigating risks and economic losses. Usually, this monitoring is done by alarms, sensors, and systems within the pipelines' range, which indicate any depressurization, leakage, or suspicious movement near it. When any suspicious event is identified, or alarms are triggered near the pipelines, a team is called to verify the location, confirm the event, and collect data on specific areas. It is common for monitoring systems along the pipelines to trigger many alerts in a short period. For this reason, the company must identify the alerts that are more critical and more likely to be confirmed as illegal tapping as quickly as possible.

Some studies focus on machine learning to detect and locate leaks (Araujo et al. 2018; Oh et al. 2018a; Rashid et al. 2014; Ribeiro et al. 2020). Instead, this study used these techniques to identify which alerts are more likely to become a fuel theft event. Therefore, this work aims to perform a predictive model capable of (1) detecting the probability of an event resulting in an illegal tapping; and (2) identifying the factors that influence the occurrence. Principles of data science and machine learning techniques were used, such as Logistic Regression and Random Forest. Applying these techniques will help optimize the company's handling of these suspected fuel theft events, avoiding sending unnecessary patrols to the site. Most importantly, the model aims to assist the decision-making and the alerts' correct prioritization.

The remainder of this work is organized as follows. The next section provides a literature review of fuel theft literature and data science techniques. Then, we present the methodology for data analysis and model development, followed by the results and discussion. Finally, we summarize the findings, conclusions, and limitations.

Literature Review

Due to the impact of the fuel theft problem, the constant monitoring of oil pipelines is critical. Sensors inside the oil pipelines can identify oil leaks or suspicious events and send out alarms. Extracting information from this data with data science techniques can be an essential tool to rapidly identify the highest priority events.

Machine learning has been widely used to detect leaks or defects in pipelines. Cruz et al. (2020) propose a technique that uses acoustic sensors with machine learning algorithms to identify and locate leakages in low-pressure gas pipelines. Kim et al. (2021) used deep neural network methods to train subsea gas pipelines' dynamic leak detection model. Rahmati et al. (2017) proposed a neural network–based method for leakage detection of a gas pipeline by using a gas flow pattern. Oh et al. (2018a) presented different machine learning algorithms, such as Random Forest and XGBoost, to detect the presence of leaks in pipelines.

The failure to detect events in pipelines carrying oil and gas was approached by Rashid et al. (2014). They presented a distributed detection system based on wireless sensor networks that identify leakage in long-range pipelines. Machine learning methods are described to improve these systems and evaluate the performance of these sensors in detecting pipeline leaks, such as Support Vector Machine (SVM) and K-Nearest Neighbor (KNN).

Oh et al. (2018b) proposed a novel machine learning method to efficiently diagnose high-pressure steam leakage in pipelines and condense features from acoustic signals. Akinsete and Oshingbesan (2019) used models to detect small leaks in a natural gas pipeline using operational parameters such as pressure, temperature, and flow rate. Such models performed better than a Real-Time Transient Model, which, according to the authors, is the industry's current preferred solution for leak detection in natural gas pipelines.

Kampelopoulos et al. (2020) compared the SVM and Decision Trees (DT) models for leak detection in noisy industrial pipelines. The data was gathered by a set of accelerometers on the surface of the pipes to measure noise during normal operating conditions and leaks. The most suitable model was the SVM. Araujo et al. (2018) and Zaoui et al. (2020) addressed the application of machine learning techniques to distributed temperature sensing (DTS), a technology commonly used for continual monitoring of leaks in pipelines.

There are also some interesting data science applications in the context of fuel theft. One example was proposed by Anifowose et al. (2012), who explored the problem of oil pipelines interdiction and the possible correlation between interdiction and essential socioeconomic variables. Obida et al. (2018) quantified the exposure of humans and the environment to oil pollution in Nigeria using advanced geostatistical techniques. One of their aims was to identify and analyze spill hotspots and the oil pipeline network. Results indicated that pipeline sabotage accounted for 66% of all oil spilled over nine years.

Fuel theft mainly occurs through the illegal tapping of oil pipelines. Thieves can divert oil or other refined products from pipelines by installing illegal taps. This practice can bring financial consequences for carriers and detrimental damage to the pipelines. For this matter, it is crucial to understand the factors that influence the most in pipeline tapping. Hasan (2016) addresses this problem and proposes a risk-based approach that uses an analytic hierarchy process (AHP) model to determine the probability of sabotage in different pipeline sections. Their findings show that pipelines located around areas with a high crime rate, sparsely populated vicinity, risk of landowner involvement, far-off police station, and leased/government land presence are more likely to suffer from sabotage.

The problem of illegal tapping was also addressed by Exaudi et al. (2018), who aimed to detect leakage of pipes by illegal tapping and proposed pressure sensors for automatic detection. They developed a leaking pipeline monitoring system using two pressure sensors in real-time based on the pressure value changes. Ribeiro et al. (2020) proposed a dynamic classifier selection technique on multiple criteria, named overall local class-specific accuracy (OLCA), to detect illegal tapping on pipelines. This system can reduce the number of false positives and FNs when dynamically selecting the classifiers of a baseline Random Forest ensemble.

Table 1 summarizes the works that used machine learning to detect pipeline leaks (or fuel theft) and illegal tapping. To the best of our knowledge, there are several studies on using machine learning to detect and locate leaks along oil pipelines. However, no studies applied these techniques to identify more likely alerts to become a fuel theft event. Similar to Ribeiro et al. (2020), this work focuses on the context of illegal tapping in pipelines. However, we propose using machine learning techniques to perform a predictive model capable of forecasting the probability of an alert that will result in an illegal tapping. Moreover, like Hasan (2016), we also focus on understanding the factors that influence the occurrence of this tapping in a real oil and gas transportation company.

Methodology

This section describes the learning process step by step. The framework used to develop the predictive models herein, adapted by Dantas (2020), is presented in Fig. 1.

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Table 1. Summary of works that used machine learning in the context of pipeline leaks or fuel theft

Authors	Year	Problem	Country	Type	Techniques
Rashid et al.	2014	Pipeline leak	Pakistan	Application	GMM; SVM; KNN
Rahmati et al.	2017	Pipeline leak	Iran	Application	ANN
Araujo et al.	2018	Pipeline leak	US	Conceptual	SVM; KNN; and others
Oh et al.	2018b	Pipeline leak	Korea	Application	RF; XGBoost; SVM; KNN; LDA
Oh et al.	2018a	Pipeline leak	Korea	Application	SVM; KNN; RF; XGBoost; C5.0Tree
Akinsete and Oshingbesan	2019	Pipeline leak	Nigeria	Application	GB; DT; RF; SVM; ANN
Cruz et al.	2020	Pipeline leak	Brazil	Application	LR; KNN; SVM-L; SVM-RBF; RF; Adaboost; XGBoost
Kampelopoulos et al.	2020	Pipeline leak	Greece	Conceptual	SVM; DT
Ribeiro et al.	2020	Illegal tapping	Brazil	Application	RF
Kim et al.	2021	Pipeline leak	Korea	Application	Deep neural networks

Note: ANN = artificial neural networks; DT = decision trees; GMM = Gaussian mixture model; GB = gradient boosting; KNN = K-nearest neighbor; LDA = linear discriminant analysis; LR = logistic regression; RF = random forest; SVM = support vector machine; SVM-L = support vector machines with linear Kernel; and SVM-RBF = support vector machine with radial basis Kernel.

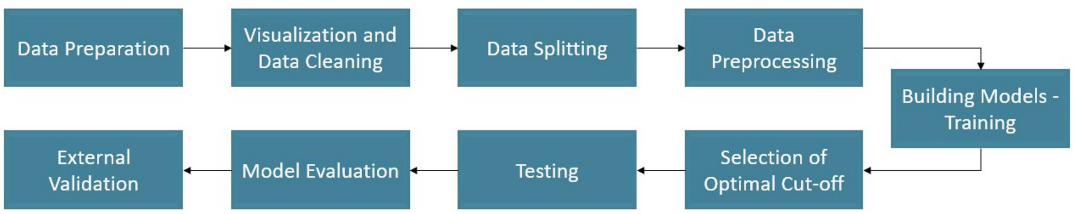

Fig. 1. Learning process framework. (Reprinted with permission from Dantas 2020.)

The process began with the extraction of the databases from the company, followed by the structuration of the data, feature engineering, missing data analysis, and outlier detection. Then, the data was split into training and test and the training data pre-processed. The models were built using supervised learning techniques based on different algorithms through hyperparameter tuning and cross-validation. The model that performed best in the training step was used to find the optimal cutoff value based on the True Positive and False Positive Rates. Once the weights were chosen, the models were tested and evaluated through performance metrics. The one with the best accuracy and specificity would be selected. Lastly, the best model was used in external validation to analyze the model's performance on new data. Appendix I provides a detailed version of the methodology described. The following subsections explain each step applied in detail for our problem.

Study Overview

Transpetro is a Brazilian company that ensures the transportation of products (such as oil, natural gas, and its derivatives) between oil-producing regions, refineries, and distribution bases. It supervises different alarm monitoring systems and receives information about alarms triggered along the oil pipelines, which can indicate leaks or damage to the pipelines caused by illegal tapping. From these alarms, verification events arise, and patrols are sent to inspect the location. The operations team needs to know which alarms have the highest priority so that patrols reach the site in the shortest time possible. The decision of which alarms are more critical is usually based on the analyst's knowledge and alerts most likely to configure an illegal tapping. In addition to avoiding illegal tapping, the correct prioritization of the alerts can reduce the number of patrols sent to inspect them, which is an advantage from an operational perspective. Prioritization is also important when there are not enough patrols to be sent to all locations. The most critical signals must be identified and assigned to the highest priority.

There are several types of monitoring systems that use different types of technologies. One of the systems used by the company uses a Negative Pressure Wave (NPW) detector, which detects leaks through pressure variations inside the pipeline. Another system uses Distributed Acoustic Detection (DAS) technology, a fully distributed sensor capable of detecting perturbations along the pipeline, along with a Negative Pressure Pulse (NPP) detector that responds when the pipeline's integrity fails and, due to its sensitivity, can detect product leaks. This detection is based on the speed of sound, which can vary according to the asset transported in each pipeline. Digging activities, for example, that might damage the pipeline, creates strong signals that are quickly detected by this sensor (Rowen et al. 2020).

In this study, our focus is detecting illegal tappings caused by perturbations in the pipeline, such as digging, before they escalate to leakage. For this reason, we focus on an alarm monitoring system called "*Sistema de Monitoramento de Faixas*" (SIMF), composed of acoustic sensors that capture movement and vibration. The alerts were restricted to digging and mechanical digging alerts, which suggests an attempt at pipeline damage to extract fuel.

Database Settings

The main database used in this work was the "Alarm Monitoring Database," which had alarms data from the SIMF monitoring system. This database has thousands of information about the triggered alarms, such as the start and end time of the alarm, duration, latitude, and longitude of the location, the name of the pipeline where it happened, the alert level (priority), and the classification of the alarms (such as "digging" and "mechanical digging"). This database had approximately 215,000 registers from January 2019 to May 2021.

Data Preparation

Feature Engineering was applied to create two new numerical features from the existing variables, the "History of illegal tapping"

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and the "Number of events nearby." The first one calculates the number of illegal tappings that occurred up to one month before the alert's date. The second one calculates the number of events that occurred within seven days before the date of the alert and within five approximate channels. The channel is the identification number of the mileage where it was triggered. Both variables consider that the illegal tapping or the event had to occur in the same alert pipeline.

Two other features were considered variables for our problem: "Duration," which calculates the duration of the alerts in minutes, and "Hour of the day," which indicates the day of the sign that was triggered, based on the hour it started. The "Hour of the day" variable was created, grouping the hours into four time ranges: Dawn, Morning, Afternoon, and Night. Each one represents a time span of 6 h, beginning at midnight (0 h). These four features ("History of illegal tapping," "Number of events nearby", "Duration," and "Hour of the day") are the explanatory variables of our model. In this problem, the dependent variable is the alert's association (or lack thereof) with an illegal tapping occurrence. The association represents the positive class, whereas the nonassociation condition means negative.

Visualization and Data Cleaning

A descriptive statistical analysis between the positive and negative classes was performed. Continuous variables were compared using Student's t-test, and categorical variables were compared using the chi-square test. The former tests the null hypothesis that the mean of the positive and negative classes is statistically identical. The latter tests the null hypothesis that the frequencies of each category for the positive and negative classes are statistically the same. The distribution of alerts and illegal tapping over time were also visualized using a line plot.

Outlier detection and treatment did not identify any abnormal values in this work. Also, there were no missing values in the variables used.

Data Splitting

The data were divided into a training set, composed of 80% of the data and a testing set, made up of the remaining 20%.

Data Preprocessing

The correlation between the variables was analyzed, dummy variables were created, and the variables were normalized (if necessary). The preprocessing must be done separately for the training and testing sets.

The Pearson method, the F statistic, and Omega-square were employed to analyze the correlation between the variables. The former was used for correlation between numerical variables, while the latter was used for categorical and numerical variables.

Normalization of numerical features to the range [0,1] was necessary before following given procedures, such as XGBoost. Some methods also required the conversion of categorical variables into binary dummy variables. Each category will become one binary variable, indicating whether the observation belongs to that category.

Building Models—Training

The following classification methods were applied: Logistic Regression; Random Forest; XGBoost; CatBoost; and Multilayer Perceptron (MLP). Moreover, a tuned grid search was used to determine the sets of hyperparameters that optimize each model fit. Hyperparameters are model parameters that define their behavior and must be outlined before we train the final model. The first method used does not have hyperparameters, but the remaining does.

Logistic regression is an example of a linear classification model used when the data analysis describes the relationship between the response variable and one or more explanatory variables. It was selected due to its simplicity and straightforward interpretability since making inferential statements about model terms is easy (Kuhn and Johnson 2013). It takes the form of Eq. (1)

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) = \alpha + \beta^{\mathrm{T}}\mathrm{x} \tag{1}$$

where α is the intercept parameter, x is a column vector of M independent variables, and β^{T} contains the coefficients of the variables, which express the relationship between the variables and the reference level (Hosmer and Lemeshow 2000). The likelihood of an event or a specific class is represented by p, and the term $(p/(1-p))$ is known as the Odds Ratio.

Tree-based classification methods segment the predictor space into several simple regions according to a set of splitting rules that can be summarized in a tree. Ensemble methods, such as Random Forests and Boosting, use trees as building blocks to create more powerful prediction models (James et al. 2013).

Random Forest combines random decision trees with Bagging to achieve high classification accuracy, outperforming other learning methods (Sagi and Rokach 2018). This model has desirable characteristics such as being robust to outliers and noise, faster than bagging or boosting, and easily parallelized (Breiman 2001). The Gini Index was used as the classification criteria to measure the quality of a split, given by Eq. (2)

$$\text{GINI}(t) = 1 - \sum_j [p(j|t)]^2 \tag{2}$$

where $p(j|t)$ is the relative frequency of class j in node t.

Boosting is an ensemble technique that tries to create a robust classifier from weaker classifiers (Galar et al. 2012). It constructs a model from the data and creates more models that attempt to correct the prior model's mistakes until it reaches a stopping criterion. XGBoost, a scalable machine learning system for tree boosting, was selected for presenting advantages such as its scalability and capacity to run faster than existing solutions (Chen and Guestrin 2016). Catboost was selected for outperforming other gradient boosting algorithms with decision trees as base predictors (Prokhorenkova et al. 2018).

Lastly, a neural network model was selected due to advantages such as high tolerance for noisy data and their ability to classify patterns on which they have not been trained (Han et al. 2011). The Multilayer Perceptron has one or more inputs, a bias, an activation function, and an output. The net is calculated by Eq. (3)

$$\text{net}_j = \sum x_i w_{ij} + \theta_j \tag{3}$$

where x_i is the i-th input, w_{ij} is the weight of input i on neuron j, and θ_j is the bias of neuron j.

A K-fold cross-validation technique was used to select the best hyperparameters, avoiding overfitting and ensuring that the model can be generalized. It divides the data set into k subsets. Splitting the training data is repeated k times. The model is trained in all subsets except one, (k-1), which is used to validate the model according to a pre-defined metric. In the end, the average of the metrics is calculated for each k model. Five folds were chosen for the

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cross-validation technique to compute the best combination of hyperparameters. Once hyperparameters were selected, the models were trained using the complete training set.

The Area Under the Curve (AUC) metric from the ROC curve was adopted to evaluate each hyperparameter combination. The ROC curve plots the True Positive Rate versus the False Positive Rate at different classification thresholds. This method checks all possible thresholds, and on an excellent classifier, the ROC curve should be as near the top-left as possible. The hyperparameters combination that resulted in the highest AUC was selected for each model.

Selection of Optimal Cutoff

The measures used to evaluate a model's performance are based on a certain threshold applied to the output scores of the classifier. The cutoff point is the probability that the model will categorize the positive or negative classes. For example, a cutoff of 0.5 means that the model will classify any observations with a probability higher than 50% as positive (otherwise, negative). An issue with this approach is that the built-in threshold applied to the scores might not be suitable, especially for unbalanced data, causing low-performance values concerning those criteria (Vanhoeyveld and Martens 2018).

On the strength of discussions with the company and the aim of the work, the main concern for the predictive model was to minimize the number of false negatives (FNs). These cases consist of situations in which the model would classify an alert as not associated with an illegal tapping when in fact, it was, which means that the alert would not be correctly prioritized, leading to a potential occurrence of fuel theft. For this reason, this was considered when determining the optimal cutoff threshold for the model. The optimal cutoff value was selected based on the Youden Index statistics provided by the ROC curve, giving greater weight to the FNs. The goal is to minimize the FNs without jeopardizing the other evaluation metrics. The optimal cutoff is the threshold that maximizes the distance to the diagonal line given by Eq. (4) (Youden 1950)

$$\max(w_1 \times \text{sensitivity} + w_2 \times \text{specificity}) \tag{4}$$

where the sensitivity is the True Positive Rate (TPR) and the specificity equals 1−False Positive Rate (FPR). The weight is represented by w_1 and w_2.

We varied the values of the weights from 1 to 4 with a step of 0.5 and calculated the model's evaluation metrics aiming to select the weights that presented the best results, reducing the number of FNs without greatly increasing the number of false positives, which was our goal, as mentioned previously. Finally, the cutoff threshold was calculated using the training set and based on the cross-validation technique considering a weight three times higher for false negative records.

Testing

Once the training was concluded, the final models were applied to the testing set using the threshold chosen. The algorithms' predictions are compared to the testing data set with known outcomes to establish the model's performance.

Model Evaluation

Evaluation metrics based on the confusion matrix analysis were used to evaluate the developed models. Metrics such as Accuracy, Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), Precision, Recall, and F1-Score were obtained. We focused on Accuracy and Specificity to evaluate the predictive model's performance on the testing set results. The accuracy is the default classification metric, but the specificity was also an important metric to consider since it measures the models' capacity to predict the negative class. From an operational perspective, the specificity shows how much the predictive model would reduce the on-site patrols to verify an alert. Therefore, such criteria helped to choose the final model for our prediction.

Running

The experiments were performed on an Intel® CORE i5 processor with 8 GB of RAM and Python 3.8.5, using *scikit-learn*, *pymatch*, *pandas*, *matplotlib*, and *seaborn*.

Results and Discussion

Data Preparation

We used the Event Inspection Database to identify the illegal tapping events from the SIMF monitoring system and searched for those events in the Alarm Monitoring Database. Considering only "Digging" and "Mechanical Digging" alerts with high levels, 69 alerts associated with illegal tapping were identified, which were considered the positive class. Alerts not associated were regarded as a negative class, provided they adhered to the following premises: a patrol team verified the alert, and it was not a maintenance alert. Moreover, we grouped into one same alert case the negative alerts of the same pipeline within five channels and a time difference of ±48 h.

The number of alerts not associated with an illegal tapping was considerably higher than those associated with an illegal tapping, causing an imbalance in the database. Building a predictive model with this database would likely estimate all alerts negatively. Therefore, a matched case-control study was performed to solve the imbalance problem.

The matched case-control study allowed us to find a specific and predefined number of negative alerts like each other according to pre-established characteristics for each signal from the positive class. In this study, we used a 3:1 proportion (controls: case), meaning that we matched each alert from the positive class to three alerts from the negative class. The variables used to match were the pipeline system (OPS) and the month. Three controls were selected for each case occurring in the same OPS and month. We obtained a database with 276 alerts through this technique, in which 69 are from the positive class and 207 are from the negative class. This database was used to build the predictive model. Fig. 2 presents all filters applied in the database.

Visualization and Data Cleaning

To better understand the trends of alerts and illegal tapping occurrences, the data before the application of the matched case-control study were plotted by month, as shown in Fig. 3. The blue line and the left axis represent the 4,423 negative cases portrayed in the database, that is, the alerts not associated with an illegal tapping. The orange line and the right axis represent the 69 positive cases. It is relevant to notice that the number of alerts is greater than the number of illegal tapping occurrences, showing that fuel theft is a rarer event.

We performed a descriptive statistical analysis to better understand and validate the data and variables, considering the data after the matching, with 69 alerts linked to illegal tapping but 207

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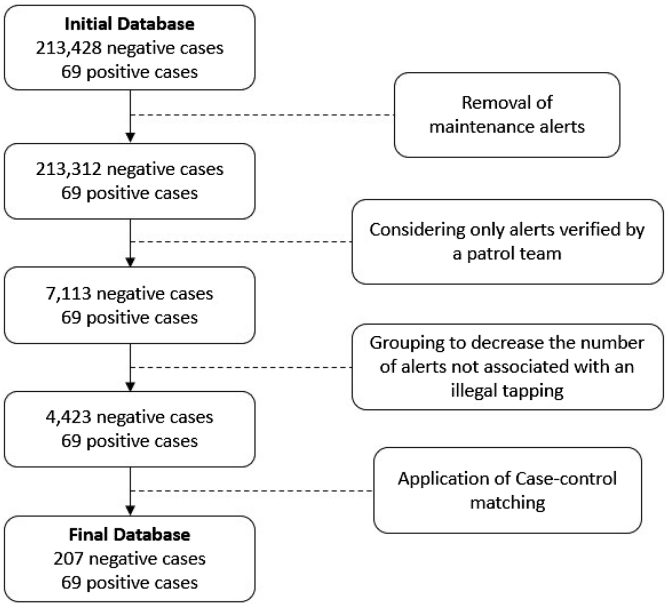

Fig. 2. Filters applied in the database.

Table 2. Descriptive statistical analysis comparing the negative and positive classes

Variables	Negative class n = 207	Positive class N = 69	p-value
	Duration (min)		
Mean (SD)	26.07 (55.33)	27.26 (30.51)	0.865
[Min, Max]	[0, 713.0]	[0, 122.0]	
	History of illegal tapping		
Mean (SD)	0.048 (0.23)	0.27 (0.51)	<0.001
[Min, Max]	[0, 2.0]	[0, 2.0]	
	Number of events nearby		
Mean (SD)	2.46 (4.80)	0.64 (1.17)	0.0019
[Min, Max]	[−1, 44.0]	[−1, 4.0]	
	Hour of the day		
Dawn	41 (19.80%)	2 (0.03%)	<0.001
Morning	44 (21.25%)	12 (17.39%)	
Afternoon	57 (25.54%)	9 (14.04%)	
Night	65 (33.41%)	46 (68.54%)	

not linked. Table 2 summarizes all four variables in the model, showing the mean, standard deviation, minimum, and maximum for each numerical variable, as well as the frequencies of each class of the categorical variable. It also presents the p-value of the t-test used for the numerical variables and the p-value of the Chi-square test used for the categorical variable.

Table 2 shows that for the "History of illegal tapping" and "Number of events nearby," we can reject the null hypothesis that the means of the negative and positive classes are statistically identical for a 95% confidence level. It implies that the variable can be significant in identifying whether the alert is associated with an illegal tapping. The p-value of the "Hour of the day" variable is also lower than 0.05, meaning that this variable can also have explanatory power. However, as for the "Duration" variable, the null hypothesis cannot be rejected.

From "History of illegal tapping," it was possible to notice that the positive class had a higher average than the negative class, indicating that if there were events of illegal tapping in the prior month at that exact location, that area should be considered more critical to inspect an alert.

Regarding the "Number of events nearby," according to the company's business rules, when there are several alerts with small durations triggered in a short period, it usually does not indicate an attempt of illegal tapping since digging is an activity that takes more time. Table 2 shows that the positive class has, on average, fewer events occurring nearby, which confirms the knowledge from the company. When analyzing the "Hour of the day" variable, we can see more alerts during the night, especially for the positive class, which is associated with illegal tapping.

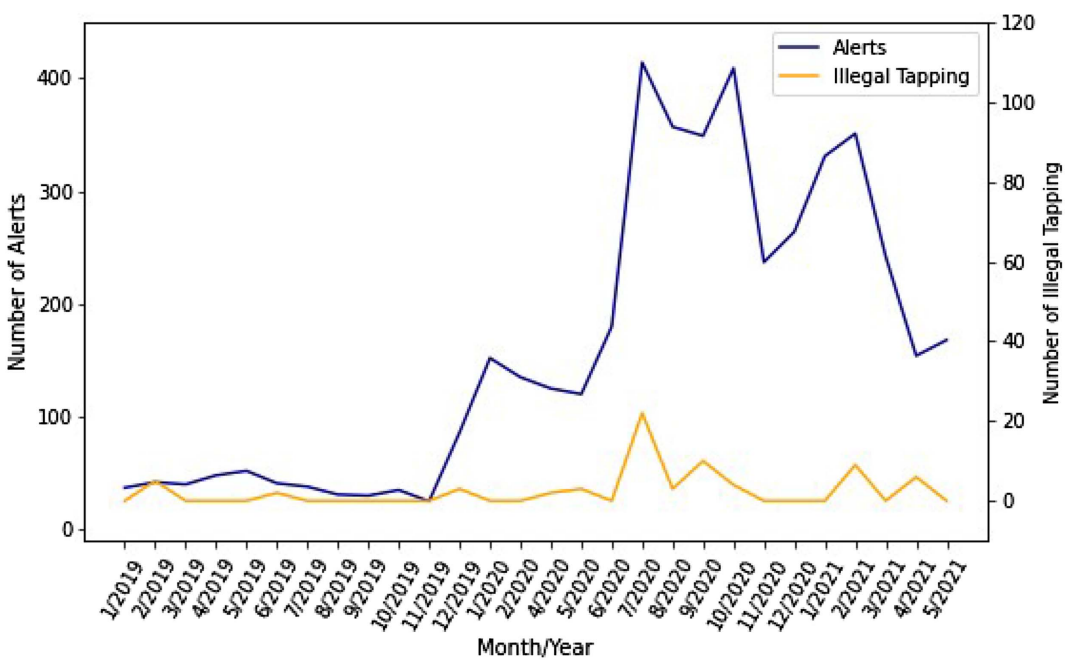

Fig. 3. Number of alerts and illegal tapping occurrences per month and year.

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Table 3. Results of the models using the testing set

Model	Cutoff	TN	FN	TP	FP	Accuracy	PPV	Sensitivity	NPV	Specificity	Precision	Recall	F1-score
Logistic regression	0.1770	19	0	15	22	0.607	0.405	1.000	1.000	0.463	0.405	1.000	0.577
Random forest	**0.2067**	**28**	**0**	**15**	**13**	**0.768**	**0.536**	**1.000**	**1.000**	**0.683**	**0.536**	**1.000**	**0.698**
XGBoost	0.1756	20	1	14	21	0.607	0.400	0.933	0.952	0.488	0.400	0.933	0.560
CatBoost	0.1613	19	1	14	22	0.589	0.389	0.933	0.950	0.463	0.389	0.933	0.549
MLP	0.0813	17	1	14	24	0.554	0.368	0.933	0.944	0.415	0.368	0.933	0.528

Note: TN = true positive; FN = false negative; TP = true positive; FP = false positive; PPV = positive predictive value; NPV = negative predictive value; and MLP = multilayer perceptron; Bold highlights the model that presented the best results.

Data Preprocessing

The data was split into training (220 alerts) and testing (56 alerts). The correlation between the chosen variables was analyzed by using the training set. This analysis is essential because variables with a high correlation could interfere with the model's results, since some models are sensitive to highly correlated variables.

The correlation analysis showed that there was not any significant correlation between the selected variables, so we used all four features as variables in our model. Appendixes II and III demonstrate the results. The "Hour of the day" variable was binarized, and the reference value was "Dawn." Moreover, the XGBoost and Catboost techniques required the normalization of the numerical variables.

Building Models and Selecting the Optimal Cutoff

After selecting the best combination of hyperparameters using the AUC value (see Appendix IV), the final models were trained. Random Forest presented the best AUC score and was chosen for the weight analysis for optimal cutoff. This analysis was performed by the Youden Index, assigning different weights for the True Positive Ratio (TPR) and False Positive Ratio (FPR). The main goal was to reduce the number of FNs without jeopardizing the model's accuracy.

Based on the sensitivity analysis results, we chose to use the weights TPR = 3 and FPR = 1 since they presented a lower number of FNs without a considerable increase in the false positives. The optimal cutoff for the Random Forest was 0.2067. It means that alerts with a probability higher than 20.67% will be considered positive, that is, an alert with a high chance of being illegal tapping. According to our model, these alerts need to be prioritized for the sending of patrols on-site to investigate.

The weight 3:1 was replicated for each model, aiming to find their optimal cutoff. The results were the following: Logistic Regression [cutoff = 0.1769]; XGBoost [cutoff = 0.1755]; CatBoost [cutoff = 0.1613]; and MLP [cutoff = 0.0813 (see Table 3)].

Model Evaluation

The next step is to test the trained models with their chosen cutoffs through the testing set. The models were evaluated by the metrics presented in the "Methodology" section.

Table 3 shows that the Random Forest model presented the best accuracy (0.768), and the best specificity (0.683) was able to correctly predict all the alerts associated with illegal tapping to this data set (FN = 0 and sensitivity = 1). Therefore, this model was selected as the final predictive model for this study.

Importance and Interpretation of the Variables

The Mean Decrease in Impurity (MDI) technique was used with the Random Forest algorithm to understand the importance of the model's variables. The categorical variables were analyzed separately, except one linearly dependent on the rest. Fig. 4 shows the importance of each variable for the Random Forest model.

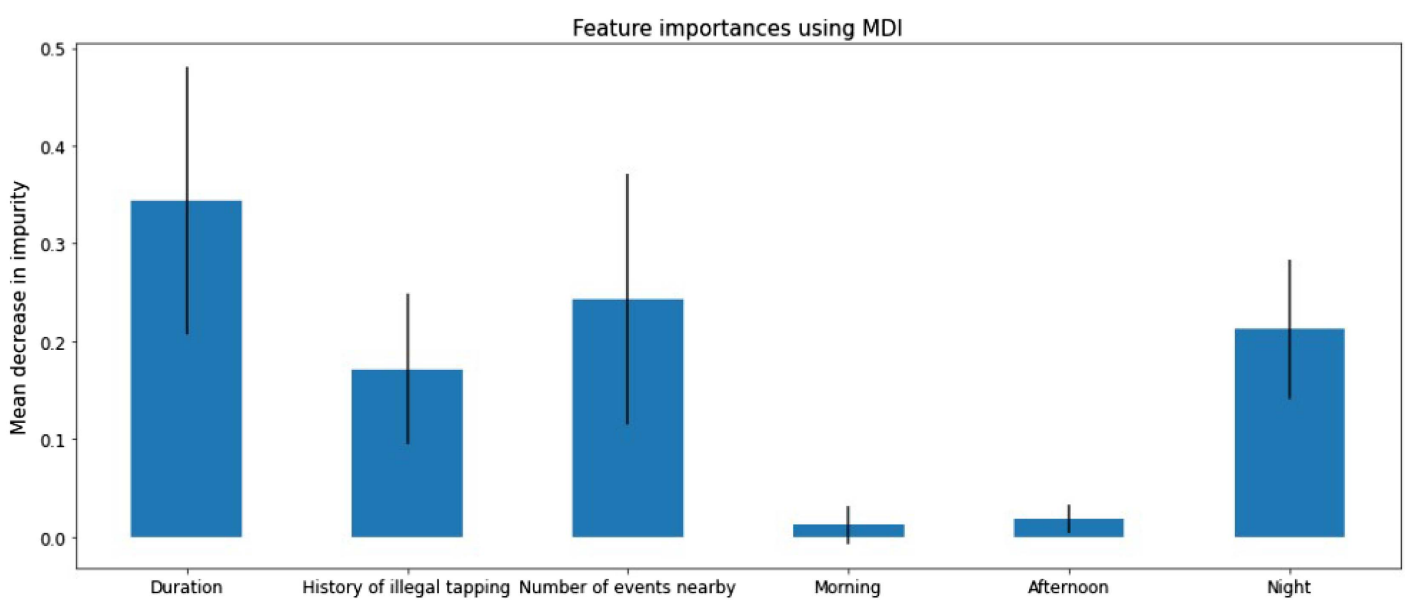

Fig. 4. Importance of each variable based on the MDI technique.

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Table 4. Results of the multiple logistic regression

Variables	Coefficient	OR
Duration	0.001	1.001
History of illegal tapping	1.461	4.311
Number of events nearby	−0.312	0.732
Hour of the day_Morning	2.134	8.445
Hour of the day_Afternoon	2.065	7.883
Hour of the day_Night	3.255	25.913

Note: OR = Odds Ratio.

The numerical variables and the "Night" category are highly significant in our predictive model.

Despite not presenting the best performance, and consequently not acting as the final predictive model, the Logistic Regression is still particularly useful due to its straightforward interpretation. Table 4 shows the coefficients and Odds Ratio (OR) for each variable using multiple logistic regression. Through ORs, we can understand the association of each variable with the response variable of an alert being associated with an illegal tapping.

According to Table 4, the variables can be interpreted as follows: OR < 1 predicts a lower chance of the alert's being associated with an illegal tapping than its reference level, whereas OR > 1 indicates a higher probability. Thus, for each additional "History of illegal tapping," the alert has a 4.31 times higher chance of being positive. On the other hand, the "Number of events nearby" variable has the opposite effect: the chance is lower for alerts with several events nearby. For the duration, each additional minute increases the likelihood of an alert's being associated with an illegal tapping by 1.001. The "Night" category is the one most likely to have alerts related to an illegal tapping (25 times higher than the reference level "Dawn"). It is worth noticing that, despite a low coefficient, the "duration" variable is an important variable to the model (Fig. 4), and models trained without this variable did not perform well.

A univariate analysis was also performed to understand if these four variables can explain an illegal tapping alone. It concluded that no single variable had a predictive capacity in the model.

External Validation

The external validation of the final predictive model used data collected between June 2021 and August 2021. There were about 50,500 new alerts in the Alarm Monitoring Database, with four new illegal tappings. The same steps of the data preparation and preprocessing, including linking all the databases, were applied to the new data.

After data preparation, the number of new registers from the Alarm Monitoring Database dropped to 841 alerts (4 positive and 837 negative). Thus, the predictive model was validated using the optimal cutoff of 0.2067 and the Random Forest model.

From the four registers of illegal tapping, three were correctly classified by the model ("TP"), while one was labeled as negative ("FN"). Out of the remaining alerts not associated with an illegal tapping, 513 alerts were correctly classified as negative ("TN"), whereas 324 were incorrectly classified as positive ("FP"). The model's accuracy was approximately 61%, as well as its specificity. The latter means that the model can reduce by 61% the sending of patrols on-site to inspect the alerts or their prioritization. The model also presented a good sensitivity (75%), which is the ability to detect positive cases.

The result of the accuracy of 61% was obtained due to the prioritization of minimizing FNs. Thus, assigning a greater weight to FNs impacted the increase of false positives in the result, generating a low accuracy. To compare the results, we used a cutoff of 0.5 for the classification of the alerts. In this case, out of the four registers of illegal tapping, only one was correctly classified by the model ("TP"), and the other three were labeled as negative ("FN"). From the alerts not associated with an illegal tapping, 747 alerts were correctly classified as negative ("TN"), and only 90 were incorrectly classified as positive ("FP"). This model's accuracy and specificity were approximately 89%, but its sensitivity was 25%.

Comparing the results of the external validation using the optimal and the 0.5 cutoffs show us that the goal of minimizing the number of FNs impacted the results. Although using a cutoff of 0.5 generated better accuracy and specificity, this model failed to detect most of the positive classifications (3 "FN" and only 1 "TP"), which is more critical in this problem since it means not associating an alert with an illegal tapping when it was an illegal topping. Using an optimal cutoff resulted in a lower accuracy, but a better sensitivity (75%) since our model was able to classify most of the positive cases correctly (1 "FN" and 3 "TP"). It provided better results considering our main objective, reducing the number of FNs. The prioritization decision will depend on the company and the impact this can have on their business, the environment, and the community.

Conclusions

A predictive model capable of forecasting the probability of an alert resulting in an illegal tapping was developed considering four variables: "duration"; "history of illegal tapping"; "number of events nearby"; and "hour of the day." The technique that provided the best results was the Random Forest, similar to Ribeiro et al. (2020). When applied to the test results, the model classified the alerts into alerts associated (or not associated) with an illegal tapping with an accuracy of 76.8%, specificity of 68.3%, and sensitivity of 100%, the last one meaning that the model can appropriately predict the positive cases. The specificity of this problem means that it is possible to reduce the sending of patrols on-site in 68.3% of cases. As for the external validation, the model also showed good performance, with an accuracy and specificity of 61%, and sensitivity of 75%.

According to our findings, the higher the alert duration, the more likely it is associated with an illegal tapping. Alerts triggered at night also have a higher probability of being associated with an illegal tapping. Conversely, an increased number of events nearby indicates a lower chance.

Transpetro uses the developed predictive model, contributing to the reduction of on-site patrols, helping analysts to prioritize alerts, and improving the efficiency of patrol responses. It also enables further improvements to combat and reduce fuel theft. From the social perspective, it promotes enhanced safety in the communities near the pipeline and a reduction in environmental problems. Among the contributions to the literature, innovations in the area stand out with the employment of machine learning to predict the probability of an event resulting in an illegal tapping.

The limitation of this study lies in the lack of integration between the databases used, which required us to perform them manually. For future research, we propose the application of balancing strategies to address the problem. We also suggest testing other machine learning techniques, such as Support Vector Machines.

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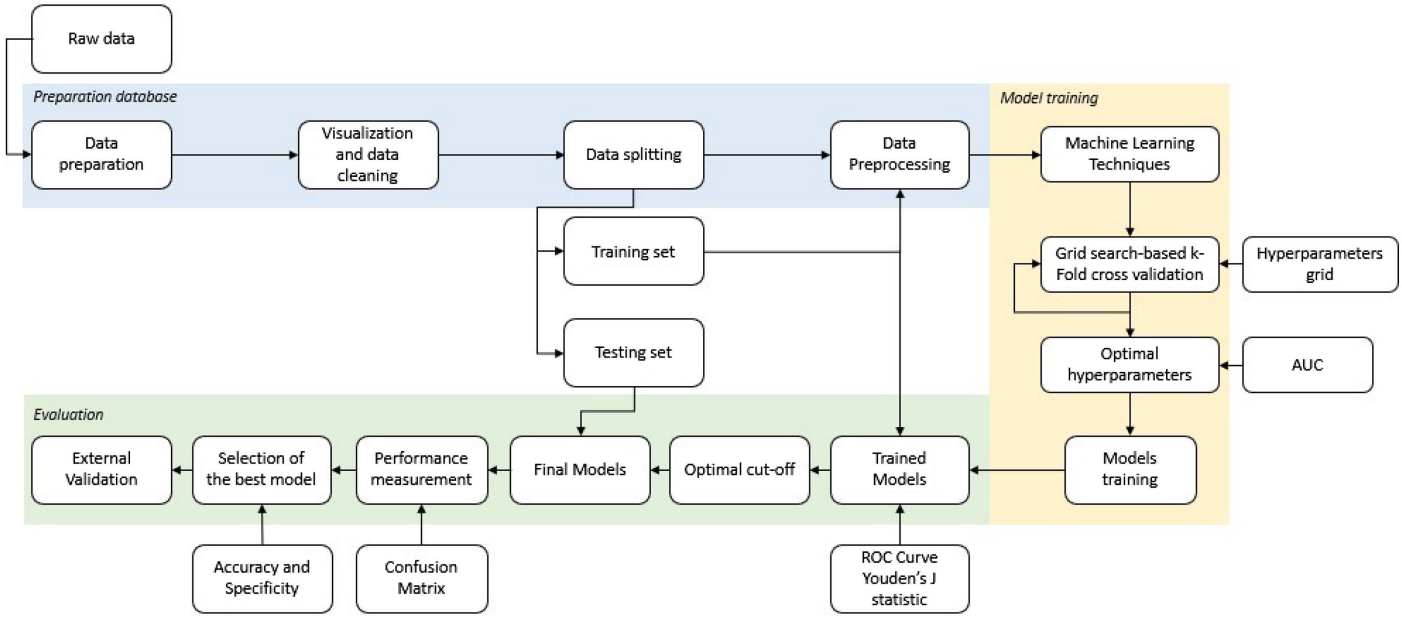

Fig. 5. Flowchart of the model building and evaluation process. (Reprinted with permission from Dantas 2020.)

Appendix I. Detailed Methodology

Fig. 5 summarizes the data preparation, building, and model evaluation process flow. After the databases' extraction, data preparation, visualization, and cleaning are done. Then, the data is split into training and testing—only the training data is pre-processed. The models are built using supervised machine learning techniques based on different algorithms through grid search-based k-Fold cross-validation and hyperparameter tuning. The metric used in the cross-validation is the AUC. The model that performed best in the training step is used to find the optimal cutoff value based on Youden's J statistic. The test set is used in the final models to evaluate their performances. The Confusion Matrix is calculated for each model's result, and then the model with the best accuracy and specificity is selected. Lastly, we use the chosen (best) model to evaluate its performance on new data—external validation.

Appendix II. Correlation between the Numerical Variables

Variables	Duration	Number of events nearby	History of illegal tapping
Duration	1.000	0.063	0.014
Number of events nearby	0.063	1.000	−0.018
History of illegal tapping	0.014	−0.018	1.000

Appendix III. Correlation between "Hour of the Day" and the Numerical Variables

Variables	F-stats	P-value	Omega-square
"Duration" and "Hour of the day"	1.730	0.161	0.008
"Number of events nearby" and "Hour of the day"	2.239	0.084	0.013
"History of illegal tapping" and "Hour of the day"	1.257	0.289	0.003

Appendix IV. Best Score and Hyperparameter for Each Algorithm

Technique	AUC	Best hyperparameters
Logistic regression	0.770	*none*
Random forest	0.822	*bootstrap [False], max_depth [10], max_features ['auto'], min_samples_leaf [4], min_samples_split [32] and n_estimators [200]*
XGBoost	0.814	*eval_metric: ['error'], tree_method: ['exact'] n_estimators [250], max_depth [5] and learning_rate [0.01]*
CatBoost	0.814	*learning rate [0.01], depth [4], iterations [200] e logging_level ['Silent']*
Multilayer perceptron	0.792	*activation: ['logistic'], batch_size: [32], early_stopping: [False], hidden_layer_sizes: [(20,)], learning_rate: ['constant'], learning_rate_init: [0.01], max_iter: [200], random_state: [10], solver: ['adam']*

Data Availability Statement

Some or all data, models, or code generated or used during the study are proprietary or confidential in nature and may only be provided with restrictions.

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Author Contributions: SH, LFD, and MVBR contributed to the study conception and design, besides data interpretation. RMV, LFD, and BB performed data processing, statistical analysis, and model development. RMV, LFD, BB, and SH drafted the first version of the manuscript. SH, MVBR, ASD, and FCR supervised the study. ASD and FCR contributed equally. All authors had full access to data, participated in data interpretation, revised the manuscript, and approved the final version of the manuscript.

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